

4c) Suppose T and S satisfy the invertibility equations (1) and (2), where T is a linear transformation. Show directly that S is a linear transformation.

Solution: Given, T is a linear transformation and $T(S(x)) = x$ and $S(T(x)) = x$ for all x in $\mathbb{R}^n$.

$\therefore T(0) = 0$ and $T(cu + dv) = cT(u) + dT(v)$ for all vectors u, v in the domain of T and all scalars c, d .

$$S(T(0)) = 0 \Rightarrow \boxed{S(0) = 0} \quad S(T(x)) = x$$

$$\Rightarrow S(T(cu + dv)) = cu + dv \Rightarrow S(cT(u) + dT(v)) = cu + dv.$$

$$\Rightarrow \boxed{S(cT(u) + dT(v)) = cS(T(u)) + dS(T(v))}$$

$\therefore$ By conditions from Sec-1.8, S is a linear transformation $\square$